

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No.....

EX-8301 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Advanced Power Electronics
(Elective - III)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is a cyclo converter?
b) What are the advantages and disadvantages of a delta connected voltage controller?
2. What are the various configurations of resonant DC power supplies? Explain with the help of circuits and waveforms.

OR

Describe SMPS using half bridge and full bridge configurations. What are the advantages and disadvantages of SMPS?

3. With the help of circuit diagram and waveforms of output voltage and current, explain working of a Buck converter feeding a resistive load. Derive the expression of the output voltage.
4. What is ac chopper? Describe the modeling of ac chopper.
5. a) Explain different types of PWM control technique for 3-phase VSI.
b) Discuss the working of 3-phase current sourced PWM CSI with appropriate diagram and waveforms.
6. How do you control the harmonics of AC chopper using symmetrical and asymmetrical waveform pattern.
7. What is the general arrangement of UPS systems? Explain different configurations of UPS working.
8. Write short notes on any two of the following:
 - a) Generalized switching cell
 - b) Space vector modulation
 - c) Design of dc inductor
 - d) Device selection in power electronics supplies

EX-8301 (GS)